

Homework (Carolina Reaper)

- Q1.** A music composition has $4 \cdot 3^2$ beats. How many beats are there in total?
 (A) 36
 (B) 48
 (C) 64
 (D) 72
 (E) 96
- Q2.** A painter's canvas has a width of $2x + 5$ feet and a length of $3x - 2$ feet. If $x = 4$, what is the area of the canvas?
 (A) 40 sq ft
 (B) 52 sq ft
 (C) 60 sq ft
 (D) 68 sq ft
 (E) 78 sq ft
- Q3.** Evaluate $2^3 + 12 \div 4 - 5$ using the correct order of operations.
 (A) 3
 (B) 5
 (C) 6
 (D) 7
 (E) 9
- Q4.** The frequency of a guitar string is given by $f = 2 \cdot 3^{x-2}$. If $x = 3$, what is the frequency?
 (A) 2
 (B) 6
 (C) 12
 (D) 18
 (E) 24
- Q5.** Simplify the expression $3(2x - 1) + 2(4x + 1)$.
 (A) $6x - 3 + 8x + 2$
 (B) $10x - 1$
 (C) $10x + 1$
 (D) $12x - 1$
 (E) $12x + 1$
- Q6.** A drum pattern has a rhythm of x beats followed by $2x + 1$ beats. If $x = 2$, how many beats are in the pattern?
 (A) 5
 (B) 7
 (C) 9
 (D) 11
 (E) 13
- Q7.** The area of a triangle is given by $A = \frac{1}{2}bh$. If the base $b = 2x + 1$ and the height $h = 3x - 2$, what is the area when $x = 3$?
 (A) 10
 (B) 12
 (C) 14
 (D) 16
 (E) 18
- Q8.** Evaluate $(2+3)^2 - 2 \cdot 5$ using the correct order of operations.
 (A) 7
 (B) 11
 (C) 15
 (D) 19
 (E) 23

Open Ended Questions (FRQ)

- Q9.** A musician is composing a piece with x instruments playing a melody and $2x + 1$ instruments playing harmony. If the total number of instruments is 21, find the value of x and the total number of beats in the composition if each instrument plays 4 beats.
- Q10.** A painter is creating a mural with a width of $x + 2$ feet and a length of $2x - 3$ feet. If the area of the mural is 50 sq ft, find the value of x and the perimeter of the mural.

Chef's Tips

- Ensure students use the correct order of operations for all numerical expressions.
- Emphasize the importance of simplifying algebraic expressions by combining like terms.
- Remind students to substitute given values into expressions carefully to evaluate results.